

Circle Equations in Context

Answer Key

High School Geometry

Quick Answers

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| 1. 49 | comparison with the boundary value = $>$ |
| 2. (a) radius = 9 km, (b) left-hand side at the farm = 81 | 6. (a) radius = 6 m, (b) right-hand side of the equation = 36 |
| 3. (a) centre = (5, 6), (b) radius = 5 m, (c) right-hand side of the equation = 25 | 7. (a) centre-radius form = $(x + 7)^2 + (y - 4)^2 - 81$, (b) radius = 9 km, (c) origin-to-centre distance = 8.06 km, — |
| 4. (a) centre-radius form = $(x - 3)^2 + (y + 5)^2 - 16$, (b) radius = 4 km | 8. (a) centre = (6, 6), (b) new radius = 7.5 m, (c) right-hand side of the equation = 56.25 |
| 5. (a) left-hand side at the ship = 288, (b) lighthouse-to-ship distance = 16.97 km, (c) | |
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What is verified

19 of 20 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problem 7. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-GPE.A.1 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 20 tagged checks.

1. Splash pad: centre (4, 7) metres, fountain at (4, 14) on the edge. Write the boundary equation.

Solution: The radius is the distance from the centre to a point on the edge, and it is easier here to build r^2 directly, since r^2 is what the equation needs:

$$r^2 = (4 - 4)^2 + (14 - 7)^2 = 0 + 49$$

Substituting $h = 4$, $k = 7$, $r^2 = 49$ into $(x - h)^2 + (y - k)^2 = r^2$ gives

$(x - 4)^2 + (y - 7)^2 = 49$

2. Cell tower: $(x + 5)^2 + (y - 2)^2 = 81$ in km. (a) Radius. (b) Left-hand side at the farm (4, 2).

Solution (a): The right-hand side of a centre-radius equation is r^2 , so $r = \sqrt{81}$.

$r = 9 \text{ km}$

(The centre is $(-5, 2)$: the form is $(x - h)^2$, so $x + 5$ means $h = -5$.)

Solution (b): Substitute $x = 4$, $y = 2$:

$$(4 + 5)^2 + (2 - 2)^2 = 81 + 0$$

so the left-hand side is

81

Because that equals $r^2 = 81$ exactly, the farm sits *on* the edge of the coverage circle.

3. Garden bed: diameter endpoints $(1, 3)$ and $(9, 9)$ metres. (a) Centre. (b) Radius. (c) Equation.

Solution (a): The centre of a circle is the midpoint of any diameter:

$$\left(\frac{1+9}{2}, \frac{3+9}{2}\right) = \left(\frac{10}{2}, \frac{12}{2}\right)$$

so the centre is

$(5, 6)$

Solution (b): The radius runs from the centre to either endpoint:

$$r = \sqrt{(9-5)^2 + (9-6)^2} = \sqrt{16+9} = \sqrt{25}$$

So

$r = 5$ m

Solution (c): With $h = 5$, $k = 6$ and $r^2 = 25$:

$$(x-5)^2 + (y-6)^2 = 25$$

Check with the other endpoint: $(1-5)^2 + (3-6)^2 = 16+9 = 25$ ✓

4. No-fly zone: $x^2 + y^2 - 6x + 10y + 18 = 0$ in km. (a) Centre-radius form. (b) Radius.

Solution (a): Group the x terms and the y terms, then complete each square. Half of -6 is -3 and $(-3)^2 = 9$; half of 10 is 5 and $5^2 = 25$.

$$\begin{aligned}(x^2 - 6x \quad) + (y^2 + 10y \quad) &= -18 \\(x^2 - 6x + 9) + (y^2 + 10y + 25) &= -18 + 9 + 25 \\(x-3)^2 + (y+5)^2 &= 16\end{aligned}$$

So

$$(x-3)^2 + (y+5)^2 = 16$$

The centre is $(3, -5)$.

Solution (b): $r = \sqrt{16}$, so

$r = 4$ km

5. Lighthouse: $(x-12)^2 + (y+5)^2 = 169$ in km, ship at $(0, 7)$. (a) Left-hand side. (b) Distance. (c) Compare.

Solution (a): Substitute $x = 0$, $y = 7$:

$$(0-12)^2 + (7+5)^2 = 144 + 144$$

So the left-hand side is

288

Solution (b): The centre is $(12, -5)$, so

$$d = \sqrt{(0-12)^2 + (7-(-5))^2} = \sqrt{288} = 16.9705\dots$$

So

$$d \approx 16.97 \text{ km}$$

Solution (c): The boundary value is 169, and 288 is larger, so the comparison is $>$

Since the left-hand side exceeds r^2 , the ship is *outside* the lighthouse's range — consistent with part (b), where $16.97 > 13$.

6. Running track: centre $(6, 2)$ metres, tangent to the line $y = -4$. (a) Radius. (b) Equation.

Solution (a): A circle tangent to a horizontal line touches it at exactly one point, directly below (or above) the centre, so the radius is the vertical gap between the centre's y -coordinate and the line:

$$r = |2 - (-4)| = |6|$$

So

$$r = 6 \text{ m}$$

Solution (b): With $h = 6$, $k = 2$ and $r^2 = 36$:

$$(x - 6)^2 + (y - 2)^2 = 36$$

Check the tangent point $(6, -4)$: $(6 - 6)^2 + (-4 - 2)^2 = 0 + 36 \checkmark$

7. Flood zone: $x^2 + y^2 + 14x - 8y - 16 = 0$ in km. (a) Centre-radius form. (b) Radius. (c) Is the origin inside?

Solution (a): Half of 14 is 7 and $7^2 = 49$; half of -8 is -4 and $(-4)^2 = 16$.

$$\begin{aligned}(x^2 + 14x + 49) + (y^2 - 8y + 16) &= 16 + 49 + 16 \\(x + 7)^2 + (y - 4)^2 &= 81\end{aligned}$$

So

$$(x + 7)^2 + (y - 4)^2 = 81$$

The centre is $(-7, 4)$.

Solution (b): $r = \sqrt{81}$, so

$$r = 9 \text{ km}$$

Solution (c): Distance from the origin to the centre:

$$d = \sqrt{(0 - (-7))^2 + (0 - 4)^2} = \sqrt{49 + 16} = \sqrt{65} = 8.0622 \dots$$

So

$$d \approx 8.06 \text{ km}$$

Explanation (c) — instructor-judged. A model response: a point lies inside a circle exactly when its distance from the centre is less than the radius. Here $8.06 < 9$, so the city centre lies inside the flood-warning zone.

Full credit: compares 8.06 with 9 and states the inside test. *Half credit:* compares the two numbers without stating the test. *No credit:* an answer read off the sign of the constant term or from a sketch.

8. Reservoir: diameter endpoints $A(2, 9)$, $B(10, 3)$ metres; rebuilt radius one and a half times as large.

Solution (a): Centre = midpoint of $\overline{AB}$:

$$\left(\frac{2 + 10}{2}, \frac{9 + 3}{2} \right) = \left(\frac{12}{2}, \frac{12}{2} \right)$$

So the centre is

$$(6, 6)$$

Solution (b): The old diameter is

$$AB = \sqrt{(10 - 2)^2 + (3 - 9)^2} = \sqrt{64 + 36} = 10,$$

so the old radius is 5 and the new radius is 1.5×5 .

7.5 m

Solution (c): The centre is unchanged and $r^2 = 7.5^2$:

$(x - 6)^2 + (y - 6)^2 = 56.25$
